

ORIGINAL ARTICLE

Effectiveness of a combined approach using PDL and Q-switched 1064 Nd: YAG laser mode for the treatment of melasma lesions in Venezuelan female patients

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ABSTRACT

Background: Melasma is a chronic hyperpigmentation disorder often presenting with a vascular component. Conventional therapies show variable efficacy and limited durability of results, highlighting the need for optimized treatment strategies.

Objective: To evaluate the effectiveness of combined laser toning (Q-switched 1064 Nd:YAG) and pulsed dye laser (PDL, Candela Vbeam) for the treatment of vascularized melasma lesions.

Methods: A total of 120 female patients aged 34–56 years with vascularized melasma underwent treatment and follow-up at baseline, 3-, 6-, and 12-months using VISIA Complexion Analysis. Standardized images (frontal, right, and left lateral) were obtained to assess percentile scores for brown spots (melasma) and red spots (telangiectatic erythema within melasma). Patients received eight weekly sessions of laser toning. Subgroups included: 26 patients treated with laser toning alone, 54 patients with PDL prior to laser toning, and 40 patients with PDL after laser toning.

Results: Median VISIA percentiles for both brown and red spots significantly increased after treatment ($p < 0.005$), reflecting a reduction of vascularized melasma lesions. Improvement was most pronounced in patients receiving PDL before laser toning ($p < 0.0001$). The therapeutic effect of PDL followed by toning persisted for 12 months, whereas patients treated with laser toning alone showed diminished effects by 6 months ($p < 0.05$).



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Conclusions: The combination of PDL followed by Q-switched 1064 Nd:YAG laser toning is a safe and effective treatment for melasma with a vascular component, providing longer-lasting results compared to laser toning alone.

Key words: Melasma, Pulsed dye laser (PDL), Q-switched Nd:YAG laser, Vascular Component, Laser Toning, Long-Term Efficacy

Introduction

Melasma is a commonly acquired hypermelanosis characterized by light to dark brown patches located on sun-exposed areas of the skin mostly over the forehead and malar areas. Although it is considered a benign skin condition, it influences a patient's quality of life. Etiological factors include genetic predisposition, exposure to UV radiation (UVR), and hormonal influence^{1,2}. Women with Fitzpatrick skin types III–V who live in tropical countries and are more exposed to UVR are particularly affected^{3–5}.

Even though hyperpigmented patches constitute the major clinical feature of melasma, the presence of telangiectatic erythema is frequent, probably as a consequence of solar elastosis induced by chronic UV exposure⁶. Histological studies have demonstrated an increase in the number as well as in the size of blood vessels within melasma lesions⁷. The vascular changes characteristic of melasma are thought to arise from the stimulation of pro-angiogenic factors such as vascular endothelial growth factor (VEGF), Stem Cell Factor (SCF), and Inducible Nitric Oxide Synthase (iNOS)⁸. An elevated expression of VEGF in keratinocytes has been suggested as an important factor for altered cutaneous vasculature in melasma^{9,10}. *In vitro* studies have demonstrated that normal human melanocytes express functional VEGF receptors¹¹. VEGF receptors in melanocytes can be up-regulated by UVR exposure¹¹ and there is evidence that VEGF receptor signaling is involved in UVR-mediated melanogenesis¹². In addition, Endothelin-1 (ET-1), which is released from endothelial cells, may also stimulate pigmentation through the activation of the endothelin receptor B at the surface of melanocytes^{8,13,14}.

Possibly, due to the complexity of the many factors influencing the appearance of melasma, to date,

no treatment has proven to be completely effective or to prevent frequent relapses. Laser toning using low-fluence 1,064 nm Q-switched Nd:YAG laser has been considered a suitable approach in recent years, offering good results^{15,16}. Nevertheless, the efficacy of the treatment is variable and is influenced by several risk factors such as hormonal contraception, thyroid disorders as well as sun exposure¹⁷.

On the other hand, Pulsed-dye laser (PDL) is considered the gold standard therapy for vascular lesions. By targeting not only melanin but also vascularization and in part elastosis, PDL has become an effective treatment option in combination with pigment-targeted modalities¹⁸. There is also evidence indicating that the use of PDL plays an important role in preventing melasma relapse^{9,19}.

Therefore, the aim of this study was to evaluate the efficacy of a combined treatment using a PDL (Candela Vbeam device) and Laser Toning (low-fluence 1,064 nm Q-switched Nd: YAG laser, Lutronic) in a group of patients exhibiting melasma with a vascular component within the lesions.

Material and methods

An observational study was carried out in an unselected group of 120 female patients (mean age: 42.5 ±16.8) attending the Laser Aesthetic Medical Center UNIMEL in Caracas from September 2015 to September 2019. The patients had clinical diagnosis of vascularized lesions of melasma. They received laser treatment using PDL (Candela Vbeam device) and laser Toning (low-fluence 1,064 nm Q-switched Nd: YAG laser, Lutronic) or laser Toning alone. Immune suppression, history of photosensitivity, collagen vascular disease, a period of ≤ 6 months after the use of

skin lightening products, pregnancy, breastfeeding and delivery date up to 6 months prior to evaluation were considered as exclusion criteria. None of the patients used depigmenting creams before or during treatment.

The study was approved by the Ethical Committee of the UNIMEL Center. Informed consent for photographic record, treatment, monitoring and publication of the data was obtained from the patients participating in the study. Digital photographic evaluations were carried out using the VISIA Complexion Analysis System (Canfield Scientific., Fairfield, NJ, USA)^{20,21} before and after three (3), six (6) and twelve (12) months after treatment. Before assessment, the facial skin of all patients was cleaned gently to remove makeup. Images were captured under standard, cross-polarized, parallel-polarized, and ultraviolet illumination, in two different close-up views (right lateral 37°, left lateral 37°). Skin phototypes (I-V) as well as the presence of brown darker spots associated with melasma lesions and red areas indicating the presence of telangiectatic erythema within the lesions were evaluated. The VISIA system allows quantitative assessment of both brown spots, indicative of melasma, and red areas, which indicates the presence of the vascular component within the lesions. Quantitative analysis of red areas and brown darker spots was performed through comparison between the individual's scores and those of people of the same sex, age and skin type provided by the VISIA database. The values range from percentile 1 to percentile 100. Location at lower percentiles corresponded to the presence of darker brown spots or red areas for each case independently.

Patients underwent face exposure with laser toning NdYAG 1064 nm, Q-switched mode (Lutronic Spectra Laser), starting with a fluence of 1.0 J/cm², increasing progressively from 1 joule up to 2 joules, using 8 mm spot size and a frequency of 5 - 10 Hz with multiple passes, in eight sessions weekly. PDL (Candela Vbeam, Fluency 6 - 9 J/cm², T: 10 ms / Spot size: 7 mm) was applied in a single session before or a week after laser toning treatment. Patients' eyes were adequately protected. After each laser session, patient faces were wiped with fresh water and sunscreen SPF30 was applied. Patients were assigned to three groups based on clinical workflow and device availability, without predefined allocation criteria: (1) laser toning alone (n = 26); (2) PDL before laser toning (n

= 54); and (3) PDL after laser toning (n = 40). All patients presented melasma with vascular involvement. The VISIA percentiles corresponding to dark brown spots indicating the presence of melasma in the face (right malar, frontal and left malar) were determined before and three (3) months after treatment. The median of the overall VISIA scores corresponding to red and dark brown spots were calculated for each patient before treatment and three (3), six (6) and twelve (12) months after treatment.

Statistical analysis was carried out using GraphPad InStat version 8.0.2 (GraphPad Inc, Ca, USA). A nonparametric 1-way ANOVA comparative study was performed. Kruskal-Wallis test was used to compare the values of the percentiles for red and brown spots according to the VISIA scale. Also, the proportion of patients with significant variations on the VISIA scale after treatment was determined. For this purpose, the difference between the percentile values before and after treatment was calculated for each patient and expressed as a percentage. An increase of $\geq 30\%$ on the VISIA percentile scale was considered a significant improvement of the melasma lesion.

Results

The study included 120 female patients, exhibiting melasma lesions with vascular compromise. According to VISIA recordings, 13% of the patients corresponded to Phototype II, 65% to Phototype III and 22 % to Phototype IV. It was observed that before treatment all patients exhibited brown spots that corresponded to values below the 70th percentile on the VISIA scale as well as red areas within the spots corresponding to values below the 55th percentile, indicating the appearance of pigmented lesions and vascular alterations compatible to the presence of melasma with telangiectatic erythema within the melasma lesions.

Figure 1 shows the effect of PDL treatment on the values of the VISIA percentiles corresponding to red areas in the group of patients who received PDL before starting laser toning treatment (n=56). It can be observed that PDL application resulted in a significant increase of patient scores on the VISIA percentile scale. Thus, the median of the percentile values obtained after treatment was significantly higher compared to

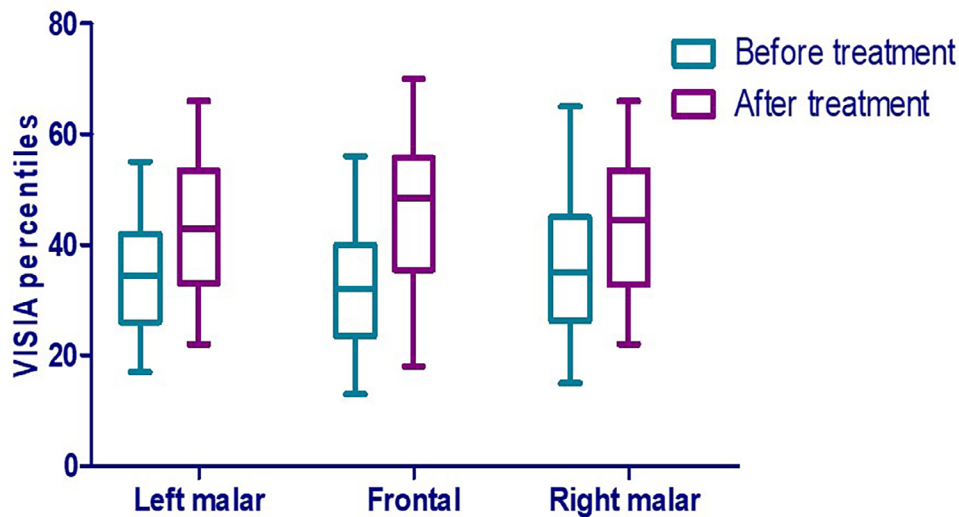


Figure 1. Effect of PDL treatment on the values of the VISIA percentiles corresponding to red areas in the group of patients who received PDL before starting laser toning treatment. Values correspond to evaluations before and three months after treatment.

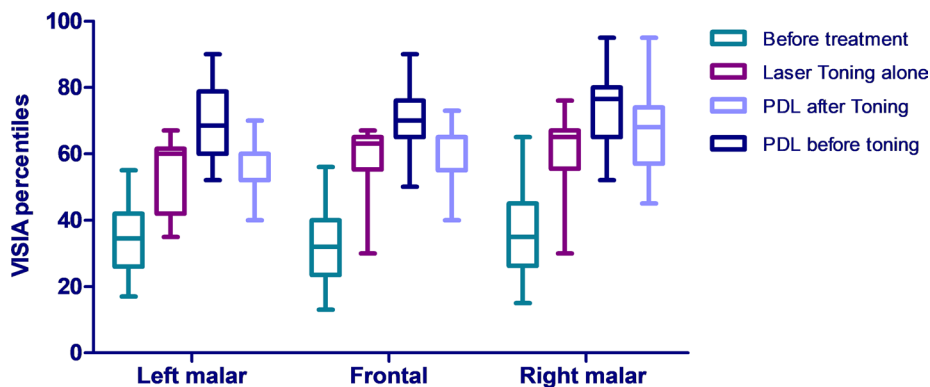


Figure 2. Effect of a combined treatment using PDL and toning in a group of Venezuelan females suffering of melasma with vascular component. The figure shows values of the VISIA percentiles for brown spots (melasma lesions) before and three months after laser toning alone or combined treatment (PDL + laser Toning).

that observed before treatment (Kruskal-Wallis statistic: 62.84; $p < 0.0001$) indicating a significant reduction of the red area and therefore of the vascular compromise within the lesions.

Figure 2 shows the values of the VISIA percentiles for brown spots before and after combined treatment (PDL + laser toning) among the different groups: Before treatment ($n=120$), laser toning alone ($n=26$), PDL applied before laser toning ($n=54$) and

PDL applied after laser toning ($n=40$). A significant improvement in the patient's scores was observed after treatment in all the groups evaluated. This was evidenced by a significant increase of the Median of the percentile values (Kruskal-Wallis statistic: 389.9; $p < 0.0001$) compared to those observed before treatment. The increase was more noticeable in the group who received PDL before toning (Dunn's post-test: $p < 0.0001$). In contrast, the values of the Median of the

percentile values observed for the group who received PDL after laser toning were similar to those observed in the group who received laser toning alone. Comparable values of the Median of the percentile values for both red areas and brown spots were observed regarding location of the melasma lesion (left malar, frontal or right malar). The proportion of patients who obtained a significant improvement ($\geq 30\%$) of their scores on the VISIA Percentile scale for brown spots (melasma lesions) was also determined. Thus, it was observed that when PDL was applied after toning, 65.5% of the patients reduced significantly the severity of their lesions in the right malar area, 64.7% in the

central facial area and 66% in the left malar area. This proportion increased markedly when PDL was applied before Toning (83% in the right malar area, 84.5% in the central facial area and 81% in the left malar area).

There was no influence of skin phototype on the proportion of patients who reduced significantly the severity of their lesions as well as on the Median of the percentile values corresponding to red areas or brown spots. Figure 3 illustrates the effect of PDL on the reduction of the red areas and of the combined treatment PDL + Toning on the improvement of the melasma lesion in a 42 year old female, patient according to the VISIA scale.

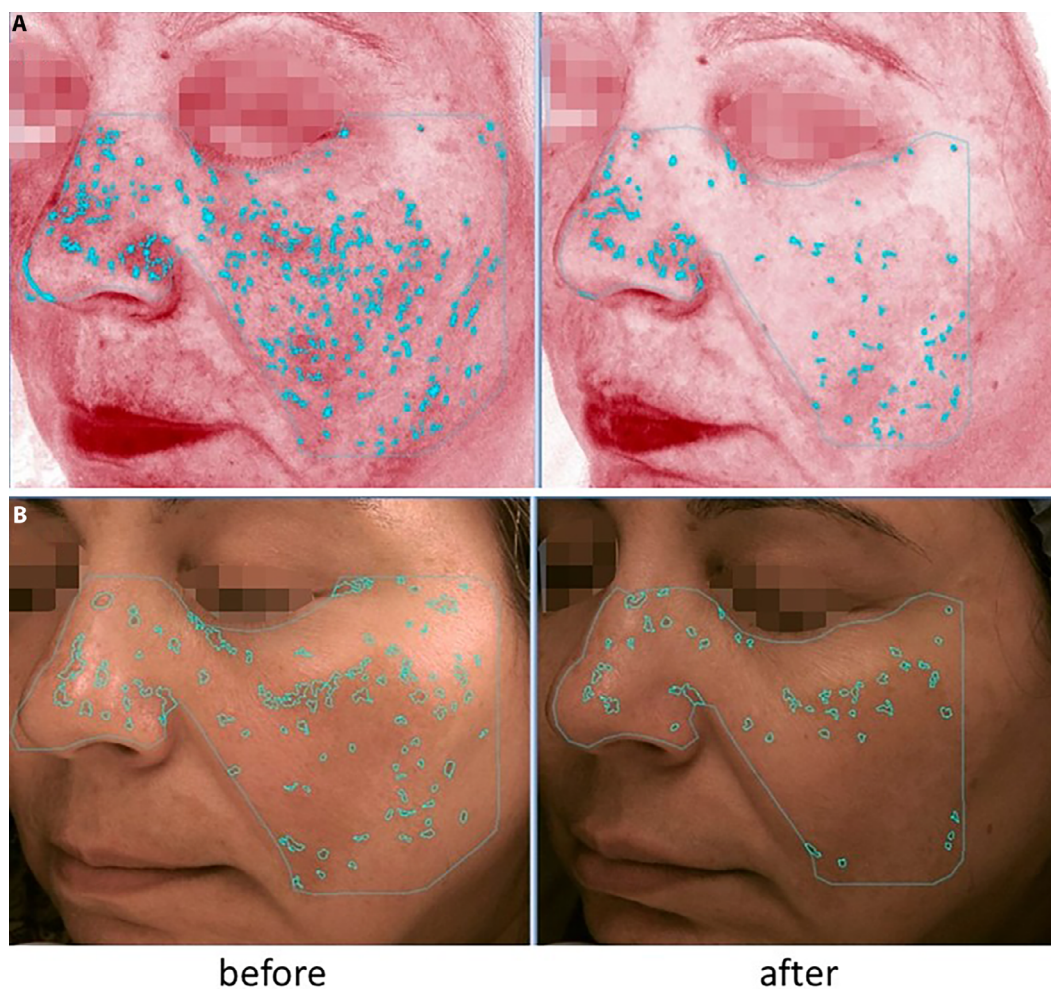


Figure 3. Female patient, 42 years of age, who experienced an improvement of 35% (according to the VISIA percentile scale) of the red area after PDL treatment (figure 3a) and of 37% (according to the VISIA percentile scale) of the melasma lesion after combined treatment: Vbeam + Toning (figure 3b). Data were recorded three months after treatment.

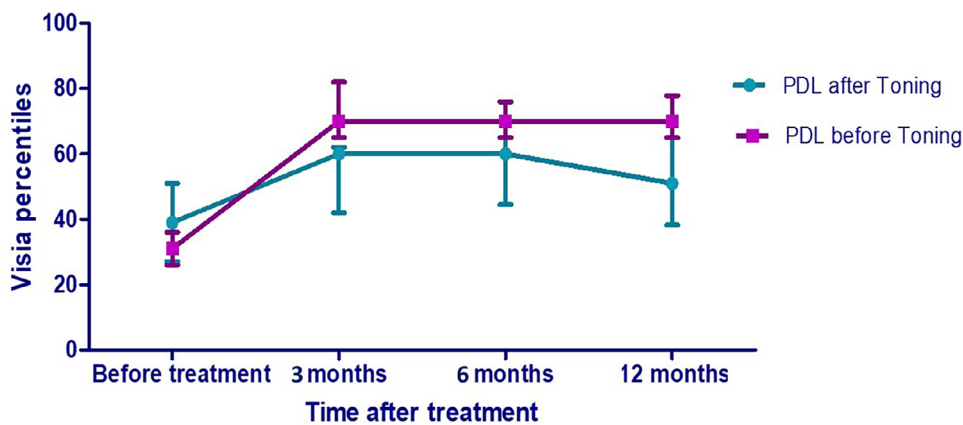


Figure 4. Over time effect of the use of PDL followed by toning to treat melasma lesions with vascular compromise.

To study differences on the effect of the use of PDL before toning with the use of PDL after toning over time, the Medians of the overall scores for dark brown spots obtained before treatment and three, six and twelve months after treatment were compared. It was observed that the values of the median of the VISIA scores increased significantly ($p < 0.005$) after treatment in both groups. These values were comparable until 12 months after treatment among those patients who received PDL followed by toning, whereas in those who received PDL after toning, the median VISIA scores significantly decreased six months after treatment ($p < 0.05$) (Figure 4). No adverse events, including blister or crust formation, hyperpigmentation, hypopigmentation, scarring, or severe inflammation, were observed. Mild erythema and oedema were reported with both laser toning and PDL (Vbeam), which resolved spontaneously within 24–48 hours. All patients expressed satisfaction with the treatment and the results achieved.

Discussion

The results of the present study confirmed that vascular targeted laser therapy improves melasma lesions when combined with laser toning. Previous studies have shown similar outcomes. For example, Cunha *et al.*²² studied six patients of refractory melasma. They were treated with Intense pulsed light (IPL) followed by toning at a monthly interval.

A significant improvement in the lesions was observed in all the patients. Another study carried out in 12 Korean patients also treated with fractionated IPL and laser toning²³, showed significant better improvement in the group treated with the combined approach than in the group treated with IPL alone.

The majority (83%) of the participants in this study exhibited the so-called “Latin type skin” corresponding to phototypes III and IV while only 17% of the patients corresponded to Phototype II. Results from previous studies have shown that individuals with darker skin types are more susceptible to developing side effects such as post-inflammatory hyperpigmentation (PIH)^{24,25} or depigmented mottling lesions²⁶. However, in the present study none of the treated patients developed side effects up to three months after the end of the treatment. Moreover, there was a noticeable improvement of the melasma lesions, regardless of skin color, among the overall group of patients. It is important to point out that the study population was instructed to use broad ultraviolet UV-spectrum sunscreens before, during and after treatment which would contribute to prevent PIH or depigmentation. Avoiding of exposure to visible light was also recommended to the patients, since shorter wavelengths of visible light can also promote hyperpigmentation^{27,28}.

In this study, the time at which PDL was applied (before or after laser toning) turned out to be a relevant factor influencing the effectiveness of the treatment. Thus, a noticeable improvement of the melasma lesions

was observed when PDL was used before laser toning while in those patients to whom PDL was applied after toning the results obtained were similar to those observed in the group receiving laser toning alone. In line with this, a previous study using Pulsed 595-nm PDL followed by the 1927-nm fractional low-powered diode laser (FDL) showed great amelioration of melasma lesions parallel to a significant improvement in erythema²⁹. In contrast, it has been reported that the use of the Q-switched 1064 Nd:YAG laser mode followed immediately by PDL results in similar values of MASI scores compared to those observed when using Q-switched 1064 Nd:YAG laser mode alone³⁰. Studies carried out in Asian patients treated with an antiangiogenetic laser have shown that improvement of dermal telangiectatic erythema within melasma lesions is due to a significant decrease in different histochemical markers for vascular antigens, including Melan-A, endothelin 1, CD34, and VEGF³¹.

The use of a combined treatment by means of Q-switched Nd:YAG and either IPL or PDL has been recommended in recalcitrant melasma with lesions involving a vascular compromise³². In line with these observations, a retrospective analysis evaluated the efficacy of sequential therapy with pulsed dye laser (595 nm) followed by low-power fractional diode laser (1927 nm) in patients with melasma lesions exhibiting subtle or subclinical telangiectatic erythema detected by spectrophotometry. More than half of the patients achieved over 50% clinical improvement, which paralleled a reduction in erythema, without evidence of rebound melasma or post-inflammatory pigmentary alterations. Moreover, patient satisfaction was consistently high. These findings reinforce the hypothesis that the vascular component plays a significant role in melasma pathogenesis, and that targeting both vascular and pigmentary elements may improve clinical outcomes while reducing recurrence risk²⁹. The results of this work showed that the effect of the combined treatment using PDL followed by toning was sustained over time as far as 12 months after treatment. In contrast, patients receiving PDL after treatment decreased significantly their VI-SIA scores values six months after treatment indicating a relapse of melasma lesions. Thus, prior inhibition of pro-inflammatory and angiogenic activity within the lesion would contribute to the effectiveness of subsequent

subcellular selective photothermolysis caused by the application of laser toning. This effect may last over time limiting the chance of relapse.

Despite the effectiveness of Laser/Light therapies to treat melasma lesions, an alternative successful option is combining laser toning with oral administration of tranexamic acid (TXA) which enhances laser therapy efficacy and decrease side effects or complications³³. TXA is an antifibrinolytic agent, inhibits UVB induced melanogenesis and neovascularization by inhibiting the plasminogen activator and plasmin activity^{12,34}. The effectiveness of such a treatment emphasizes the key role of the vascular component of melasma.

The results of this work provide evidence supporting the advantages of combining the use of PDL followed by laser toning to treat melasma lesions and highlights the importance of photoprotection on the success of the treatment. However, still more work is needed to validate the efficacy of this approach. Also, a detailed previous study of the clinical conditions of each patient is essential to select the best therapeutic option. This is particularly important in Latina women from South American countries who are more susceptible to developing melasma.

A limitation of this study is its observational design, as patients were not randomly allocated to treatment groups. Therefore, although all patients presented a vascular component of melasma, a potential selection bias may have influenced the results. The findings of this study are preliminary, and further research involving larger sample sizes is recommended to substantiate the advantages of this approach.

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Conflict of Interest: The authors declare that they have no commercial or financial relationships that could be construed as a potential conflict of interest.

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